

EXP NO:1

Transformation between Cartesian, Cylindrical and spherical coordinate system

AIM: To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required:

- SCILAB version 6.0.1

TEST CASE 1:

PRATICAL CALCUACTION:

```
EXP1.t.scilab (C:\Users\sanja\Documents\EXP1.t.scilab) - Scilab
EXP1.1.scilab X
1 clc;
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atan(y/x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=',[r pi z]);
8 r=sqrt(x^2+y^2+z^2);
9 theta=acos(z/r);
10 pi=atan(y/x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=',[r theta pi]);
12
```

Enter the value of X=-5

Enter the value of y=3

Enter the value of z=-5

"Cartesian to cylindrical coordinate system of p(r,pi,z) ="

5.8309519 -30.963757 -5.

"Cartesian to spherical coordinate system of s(r,theta,pi) ="

7.6811457 130.61286 -30.963757

exec: Wrong number of output argument(s): 0 expected.

--> |

THEROTICAL CALUCATION:

Handwritten calculations for converting Cartesian coordinates $(x, y, z) = (-5, 3, -5)$ to cylindrical and spherical coordinates.

Given: $x = -5$, $y = 3$, $z = -5$

Cylindrical coordinates (r, θ, z) :

$$r = \sqrt{x^2 + y^2} = \sqrt{(-5)^2 + (3)^2} = 5.8309519$$
$$\theta = \tan^{-1}(y/x) = \tan^{-1}(3/-5) = -30.963757$$

Spherical coordinates (ρ, ϕ, θ) :

$$\rho = \sqrt{x^2 + y^2 + z^2} = 7.6811457$$
$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{-5}{7.6811457}\right) = 130.61286$$

Final results: $(r, \theta, z) = (5.8309519, -30.963757, -5)$ and $(\rho, \phi, \theta) = (7.6811457, 130.61286, -30.963757)$

TEST CASE 2:

PRATICAL CALCULATION:

```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
r=sqrt(x^2+y^2);
pi=atan(y/x);
disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]);
r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
pi=atan(y/x);
disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]);
```

Enter the value of r=5

Enter the value of pi=60

Enter the value of z=-2

"cylindrical to cartesian coordinate system of p(x,y,z)="

2.5 4.330127 -2.

"cylindrical to spherical coordinate system of s(r,pi,theta)="

5.3851648 60. 111.80141

exec: Wrong number of output argument(s): 0 expected.

THEROTICAL CALUCATION:

$\rho = 5$ $\phi = 60^\circ$

$$x = \rho \cos \theta = 5 \cos(60^\circ) = 2.5$$
$$y = \rho \sin \theta = 5 \sin(60^\circ) = 4.330127$$
$$z = -2$$
$$r = \sqrt{x^2 + z^2} = \sqrt{(2.5)^2 + (-2)^2} = 3.201562$$
$$\theta = \cos^{-1} \left(\frac{-2}{3.201562} \right) = 111.80141^\circ$$

TEST CASE 3:

PRATICAL CALCULATION:

```
clc;
r=input('Enter the value of r=');
pi=input('Enter the value of pi=');
theta=input('Enter the value of theta=');
x=r*sind(theta)*cos(pi);
y=r*sind(theta)*sin(pi);
z=r*cosd(theta);
disp('spherical to cartesian coordinate system of p(x,y,z)=',[x y z]);
p=r*sin(theta);
pi=pi;
z=r*cos(pi);
disp('spherical to cylindrical coordinate system of s(p,pi,z)=',[p pi z]);
```

Enter the value of r=3

Enter the value of pi=40

Enter the value of theta=120

"spherical to cartesian coordinate system of p(x,y,z)="

-1.7327559 1.9358608 -1.5

"spherical to cylindrical coordinate system of s(p,pi,z)="

1.7418336 40. -2.0008142

exec: Wrong number of output argument(s): 0 expected.

THEROTICAL CALUCATION:

The image shows handwritten mathematical work on a piece of paper. At the top, there are some faint, partially visible equations: $\rho = \cos(5.3801648)$ and $\phi = 120$. The main calculations are for converting spherical coordinates (r, θ, ϕ) to Cartesian coordinates (x, y, z) . The given values are $r = 3$, $\theta = 40^\circ$, and $\phi = 120$. The calculations are as follows:

$$x = r \sin \theta \cos \phi = 3 \sin 40^\circ \cos 120 = -1.9902418$$
$$y = r \sin \theta \sin \phi = 3 \sin 40^\circ \sin 120 = 1.670012$$
$$z = r \cos \theta = 3 \cos 40 = 2.2981333$$

There are also some intermediate steps and corrections shown:

$$z = r \cos \theta = 3 \cos(40) = -1.5$$
$$\rho = r \sin \theta = 3 \sin(40) = 2.5980762$$
$$\phi = \phi = 40^\circ$$
$$z = r \cos \theta = 3 \cos(40) = 2.2981333$$

Handwritten notes include "O/p" and a signature.

RESULT:

The conversion of coordinates from Cartesian to Cylindrical and Spherical systems and vice versa was successfully performed and verified.